

Manuel Carlos Ramos Alves

SOFTWARE ENGINEER

Porto, Portugal · manuelcralves@gmail.com
linkedin.com/in/manuel-alves-a56768321 · github.com/manuelcralves

PROFILE

Software engineer with an MSc from FEUP, University of Porto, where my dissertation was developed at CERN on constraint-based planning optimization. I built and run worldcup2026ml.pt, a live machine-learning platform forecasting the FIFA World Cup.

CORE SKILLS

Languages: Python · TypeScript · JavaScript · Java · C++ · C# · SQL · PHP · C

Front-end: React · Next.js · Angular · Material UI · Bootstrap · Three.js · HTML & CSS

Back-end: Spring Boot (REST APIs, Hibernate/JPA) · Flask · Django · Express.js · Laravel

Data & ML: pandas · NumPy · scikit-learn · Google OR-Tools (CP-SAT) · Apache Solr · NLP & embeddings · LLM APIs · RAG & vector search (pgvector) · Power BI

Databases: Oracle · PostgreSQL (Supabase) · SQLite · CouchDB

DevOps & Tools: Git · Docker · GitHub Actions (CI/CD) · GitLab · Linux & Bash · Figma · Unity

PROFESSIONAL EXPERIENCE

zerozero (ZOS, Lda.) | Full-Stack Developer

Jul 2026 – Present

Vila Nova de Gaia, Portugal

CERN | MSc Thesis Student (Cooperation Associate)

Sep 2025 – Jul 2026

Geneva, Switzerland (remote phase from Porto; on-site March – May 2026)

- Developed the MSc thesis “Intelligent Planning Optimization for Programmed Stops at CERN”: state-of-the-art review on constraint programming and resource-constrained project scheduling (RCPSP), requirements gathering with CERN stakeholders, and end-to-end system architecture.
- Extended CERN’s internal Scheduling Tools platform with a React + TypeScript + Material-UI front-end (interactive Gantt chart, template engine and schedule editor with FS/SS/FF/SF dependencies and MS Project-style constraints) over a Spring Boot + Oracle back-end.
- Built a Python + Flask microservice using Google OR-Tools (CP-SAT) for constraint-based schedule optimization (working-time calendars, CERN holidays, team/location concurrency limits), with MSPDI XML export for Microsoft Project interoperability; both the front-end and the optimization service were merged into the platform in use by the EN-ACE team.

Pontual - IT Business Solutions | Software Developer Intern

Feb 2024 – Jun 2024

Santa Maria da Feira, Portugal

- Developed a procurement web portal in Angular + TypeScript + Bootstrap, integrated with the Cegid PRIMAVERA ERP for data synchronization.
- Implemented multi-level approval workflows, team management and hierarchical role-based permissions for orders and expenses, plus a dashboard landing page with interactive charts.
- Designed UI prototypes in Figma and produced flow and architecture diagrams documenting the end-to-end procurement process.

Other Experience | Part-time & seasonal alongside studies

2019 – 2026

- Sports Coordinator at the AMB Volleyball Cup (seasonal, 2019 – 2026), running on-court operations for a 6,000+ athlete tournament: team check-ins, volunteer and referee deployment, live scheduling and score reporting.
- Café server (Futpark, 2022 – 2026) and pizza-delivery driver (2021 – 2022), part-time throughout my studies; volunteer at Web Summit 2024, Lisbon.

EDUCATION

MSc in Informatics and Computing Engineering | Faculty of Engineering, University of Porto

Sep 2024 – Jul 2026

Final grade: 17/20 · Dissertation: 19/20 — “Intelligent Planning Optimization for Programmed Stops at CERN”, developed at CERN, Geneva.

BSc in Informatics and Computing Engineering | Faculty of Engineering, University of Porto

Sep 2021 – Jun 2024

Final grade: 15/20 · Capstone: procurement web portal with ERP integration, developed at Pontual.

Initially enrolled in the integrated MSc in Electrical and Computer Engineering at the same faculty (2019 – 2021) before switching to Informatics.

PROJECTS

World Cup 2026 ML Predictor | Personal project, live at worldcup2026ml.pt Jun – Jul 2026

- Built and deployed a live ML web app forecasting the FIFA World Cup 2026: Poisson + Elo model over 150 years of results, with million-scale Monte Carlo simulation for title odds, brackets and Golden Boot projections (Python, pandas, NumPy); its final predictions proved highly accurate across the tournament.
- Added “Beat the Machine”, a live prediction game scored against the model, built with Google OAuth + Supabase (PostgreSQL/RLS), a vanilla-JS dashboard and a GitHub Actions pipeline that ingests live data and redeploys every 2 hours.

NoSQL Game Store Platform | Non-Relational Databases Feb – Jun 2025

- Steam-like game store built as a full-stack app with a React/TypeScript front-end, a Flask back-end and CouchDB for document storage, featuring authentication, a Recharts analytics dashboard, Docker Compose deployment, and custom map/reduce views and aggregations across users, games, orders and carts.

Sentiment-Based Recommender & Community Forecasting | Machine Learning Complements Feb – Jun 2025

- Sentiment-enhanced recommender on Yelp data (collaborative filtering with KNN/SVD, VADER sentiment, TF-IDF embeddings), plus community-evolution forecasting using graph-based community detection and Holt-Winters time-series smoothing.

Movie Information Retrieval System | Information Processing and Retrieval Sep 2024 – Jan 2025

- Built a search engine over a custom IMDb dataset scraped and cleaned in Python; indexed in Apache Solr with basic and boosted schemas and evaluated with TREC metrics (precision/recall).
- Extended it with semantic search via sentence-transformer embeddings behind an Express.js web interface.

LANGUAGES

Portuguese	Native or bilingual proficiency
English	Full professional proficiency (C2)
Spanish	Elementary proficiency (A2)
French	Elementary proficiency (A2)

INTERESTS

Volleyball: 2× Portuguese national champion (U16 2015/16, U21 2020/21), with 10+ years of federated competition.